WASP-36b

R-1443

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-36_210223` · created 2026-08-01T08:35:42+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459268.7452 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.043
Transit depth (Rp/Rs)^2	2.72 +/- 1.43 [%]
Orbital Inclination (inc)	85.32 +/- 2.47
Airmass coefficient 1 (a1)	1.1 +/- 0.029
Airmass coefficient 2 (a2)	-0.066 +/- 0.072
Scatter in the residuals of the lightcurve fit is	2.63 %
Best Comparison Star	#3 - [299, 242]
Optimal Method	PSF photometry
Transit Duration (day)	0.086 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.043 vs 0.1368 (deviation 0.66σ)
Duration fit vs published	0.086 d vs 0.0758 d (deviation 0.73σ)
Mid-time O–C	8.2 min
Depth SNR	3.8
Residual scatter	2.63 %

Light curve

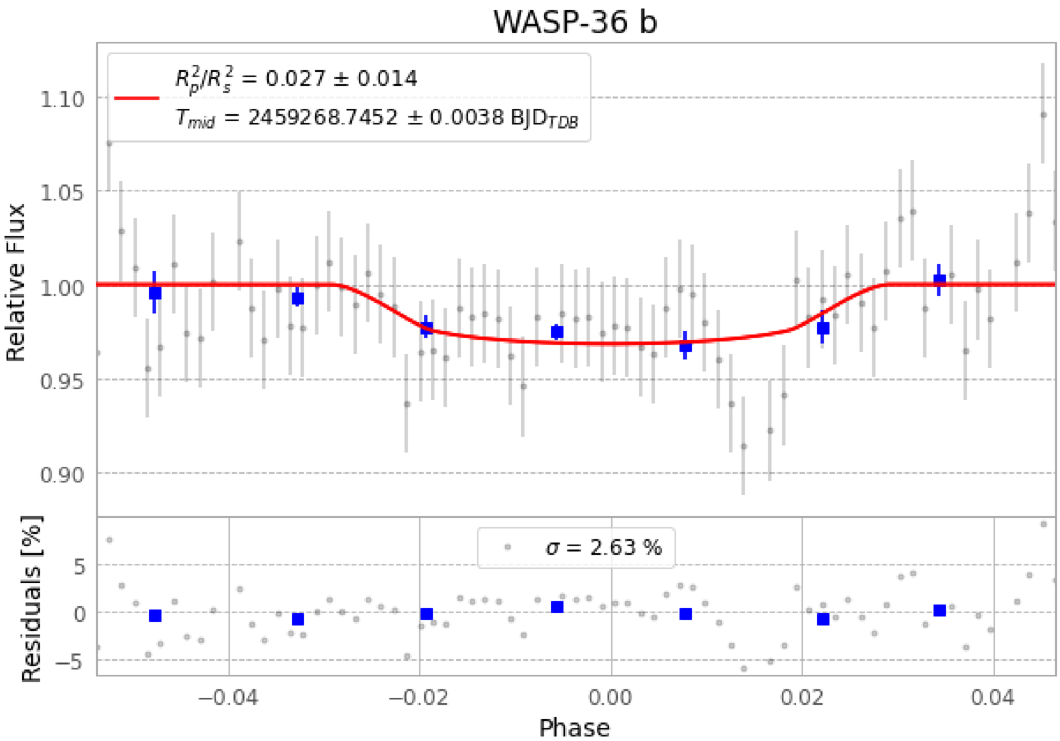

Figure 1 — FinalLightCurve_WASP-36 b_2021-02-22.png (SHA-256 43dba5eb6ecbb3aa…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [442, 225], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 fcbd0daf28445598…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

77 FITS frames (51 MB), `WASP-36210223034514.FITS` → `WASP-36210223073312.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-210223.log (f8b6bbeb74f1…), FinalLightCurve_WASP-36 b_2021-02-22.png (43dba5eb6ecb…), FinalLightCurve_WASP-36 b_2021-02-22.csv (620cb59b98fd…)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)